

## # Simulation results

\*massive-binary-sim v0.1.0 · git `UNKNOWN(no` · seed 42 · wallclock 0.07 s\*

### ## 1. Initial conditions

| quantity                                | value                    |
|-----------------------------------------|--------------------------|
| M1, M2 (ZAMS)                           | 40, 30 Msun              |
| separation / period                     | 9e+03 Rsun / 1.182e+04 d |
| eccentricity                            | 0.1                      |
| metallicity Z                           | 0.0142                   |
| spins ( $\Omega/\Omega_{\text{crit}}$ ) | 0, 0                     |

### ### Active prescriptions

| toggle             | choice          |
|--------------------|-----------------|
| gravity            | `2.5PN`         |
| wind               | `vink2001`      |
| convection         | `ledoux`        |
| overshoot          | `exponential`   |
| am_transport       | `taylor_spruit` |
| mt_stability       | `zeta`          |
| mt_rate            | `kolb_ritter`   |
| ce_formalism       | `alpha_lambda`  |
| opacity            | `kramers`       |
| reaction_rates     | `reaclib`       |
| sn_engine          | `delayed`       |
| tides              | `hut1981`       |
| structure_fidelity | `semi_analytic` |
| third_body         | `False`         |

### ## 2. Evolutionary timeline

| t [Myr] | event             | detail                                                                        |
|---------|-------------------|-------------------------------------------------------------------------------|
| 0.014   | star 1 -> MS      | reduced-model phase                                                           |
| 5.158   | star 1 -> HG      | reduced-model phase                                                           |
| 5.253   | star 1 -> CHeB    | reduced-model phase                                                           |
| 5.768   | star 1 -> preSN   | reduced-model phase                                                           |
| 5.781   | star 1 -> remnant | reduced-model phase                                                           |
| 0.014   | star 2 -> MS      | reduced-model phase                                                           |
| 6.504   | star 2 -> HG      | reduced-model phase                                                           |
| 6.631   | star 2 -> CHeB    | reduced-model phase                                                           |
| 7.322   | star 2 -> preSN   | reduced-model phase                                                           |
| 7.322   | star 2 -> remnant | reduced-model phase                                                           |
| (at SN) | **SN1**           | star 1 -> BH 33.91 Msun (delayed); bound after SN                             |
| (at SN) | **SN2**           | star 2 -> BH 18.74 Msun (delayed); bound after SN                             |
| 7.322   | **dco_formed**    | wide double compact binary (BH+BH), GW merger time 49661796998.9 Gyr > Hubble |

\*\*Fate:\*\* wide double compact binary (BH+BH), GW merger time 49661796998.9 Gyr > Hubble

### ## 3. Final-state table

| quantity    | star 1                  | star 2        |
|-------------|-------------------------|---------------|
| mass [Msun] | 33.91                   | 18.74         |
| phase       | remnant                 | remnant       |
| remnant     | BH 33.91 Msun (delayed) |               |
| remnant (2) |                         | BH 18.74 Msun |
| separation  | 1.0363e+04 Rsun         |               |

```
| period | 1.6839e+04 d | |
| eccentricity | 0.1057 | |
```

#### ## 4. Derived quantities

```
| quantity | value | unit | formula |
|---|---|---|---|
| chirp mass  $M_c$  | 21.76 |  $M_{\text{sun}}$  |  $M_c = (m_1 m_2)^{3/5} / (m_1 + m_2)^{1/5}$  |
| mass ratio  $q$  | 0.5527 | - |  $q = m_2 / m_1$  ( $\leq 1$ ) |
| symmetric mass ratio  $\eta$  | 0.2293 | - |  $\eta = m_1 m_2 / (m_1 + m_2)^2$  |
| integrated wind loss ( $\text{star1} / \text{star2}$ ) | 2.315 |  $M_{\text{sun}}$  |  $\int |\dot{M}_{\text{wind}}| dt$ ;  $\text{star2} = 9.02 M_{\text{sun}}$  |
| integrated RLOF transfer | 0 |  $M_{\text{sun}}$  |  $\int |\dot{M}_{\text{RLOF}}| dt$  (gross, donor side) |
| net  $\Delta M_1$  | 6.08 |  $M_{\text{sun}}$  |  $M_1(\text{ZAMS}) - M_1(\text{final})$  |
| net  $\Delta M_2$  | -11.25 |  $M_{\text{sun}}$  |  $M_2(\text{final}) - M_2(\text{ZAMS})$  |
| orbital separation (final) | 1.0363e+04 |  $R_{\text{sun}}$  | state variable |
| orbital period (final) | 1.6839e+04 | day |  $P = 2 \pi \sqrt{a^3 / G(m_1 + m_2)}$  |
| orbital binding energy | -1.1634e+47 | erg |  $E = -G m_1 m_2 / (2a)$  |
| orbital angular momentum | 5.3576e+55 |  $\text{g cm}^2 / \text{s}$  |  $J = \mu \sqrt{G M a (1 - e^2)}$  |
| GW merger time | 4.9662e+19 | yr | Peters 1964 eq. 5.14 (eccentric) |
| GW merger time | 4.9662e+10 | Gyr | as above |
| peak GW frequency (ISCO) | 83.5 | Hz |  $f = c^3 / (6^{3/2} \pi G M)$  |
| char. strain at ISCO ( $D=100$  Mpc) | 9.7255e-21 | - | Maggiore 2007 eq. 4.44 |
```

#### ## 5. Error budget

**\*\*Numerical error\*\*** (from the convergence study):

- not run in this invocation (`--convergence` flag)

**\*\*Physical / model uncertainty\*\*** dominates and is quantified in §6-7. The reduced single-star evolution model (fitting formulae, not a structure)

#### ## 6. Free-parameter sensitivity (one-at-a-time)

\_not run (`--sensitivity`)\_

#### ## 7. Theory comparison (competing prescriptions)

\_not run (`--theory-compare`)\_

#### ## 8. Validation suite results

```
| test | status | detail |
|---|---|---|
| kepler_closure | PASS | 200 orbits: max|dE/E|=1.77e-11, max|dL/L|=8.76e-12, |da/a|=1.52e-11 (tol 1e-06) |
| lane_emden_analytic | PASS | n=0: max|dtheta|=5.72e-16, |dxil|=8.88e-16; n=1: max|dtheta|=1.55e-14, |dxil|=1.82e-14; n=5: max|dtheta|=5.44e-14, |dxil|=1.82e-14 |
| periastron_advance_1PN | PASS | 1PN d(omega)/orbit measured=6.157175e-05 rad, predicted 6 pi GM/(c^2 a(1-e^2))=6.155112e-05, frac diff=0.034% |
| hulse_taylor_decay | PASS | dPb/dt(Peters)=-2.402102e-12 s/s vs GR prediction -2.402630e-12 (Weisberg & Huang 2016); residual 0.022%. Merger |
| energy_conservation_2PN | PASS | 2PN conservative: secular |dE/E| over 120 orbits = 1.68e-08 (tol 1e-07); max|dL/L|=1.12e-05 |
| eddington_limit_flag | PASS | 120  $M_{\text{sun}}$  ZAMS  $\Gamma_{\text{Edd}}=0.406$ ; 10x-luminosity state flags  $\Gamma > 1$ : True |
| solar_structure_eddington | PASS | Eddington model 1  $M_{\text{sun}}$ :  $P_c=1.256e+17$  (ref 2.4e17),  $T_c=1.185e+07$  (ref 1.57e7),  $\rho_c=76.9$  (ref 150),  $L=0.001$  (ref 150) |
```

#### ## 9. Assumptions & limitations

- **\*\*1-D, orbit-averaged.\*\*** No 3-D hydrodynamics; the orbit is treated by secular equations except in the dedicated Kepler-closure test.
- **\*\*Reduced stellar evolution.\*\*** Single-star tracks are fitting formulae (Hurley+ 2000-style), not Henyey structure integrations. Core masses, radii and lifetimes carry ~10-30% systematic uncertainty.
- **\*\*No magnetic fields\*\*** beyond the Tayler-Spruit AM-transport toggle.
- **\*\*No rotational mixing feedback\*\*** on the tracks (rotation only sets  $\Omega_{\text{crit}}$  and tidal rates here).
- **\*\*Mass-transfer rate\*\*** uses a Ritter/Kolb-Ritter-style analytic nozzle

- with an efficiency prefactor (KNOWN\_GAPS BIN-MDOT).
- **\*\*GW waveform\*\*** is a 0PN Newtonian chirp -- qualitative, not for matched filtering.
- **\*\*3.5PN dynamics\*\*** not implemented (KNOWN\_GAPS PN-35).
- Full list: KNOWN\_GAPS.md.

## ## 11. Bibliography

See `REFERENCES.bib` for the full list with DOIs. Key prescriptions used in this run: Peters (1964); Mora & Will (2004); Eggleton (1983); Ritte